

Wu, G.; Hu, Q.

"Key dimensions in which GenAI ChatGPT supports second language teaching and learning: A scoping literature review"

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KEY DIMENSIONS IN WHICH GENAI CHATGPT SUPPORTS SECOND LANGUAGE TEACHING AND LEARNING: A SCOPING LITERATURE REVIEW

DIMENSIONES CLAVE EN LAS QUE LA GENAI CHATGPT APOYA LA ENSEÑANZA Y EL APRENDIZAJE DE SEGUNDAS LENGUAS: UNA REVISIÓN DE ALCANCE DE LA BIBLIOGRAFÍA

WU, Guowei

University of Malaga

guoweiwu67@gmail.com

HU, Qiuxiang

University of Malaga

qiuxianghu6@gmail.com

Abstract

In recent years, AI has advanced rapidly, showing notable potential in foreign language teaching and learning. Yet, as an emerging technology, its educational integration remains exploratory. Most studies focus on isolated functionalities or small-scale findings, lacking robust theoretical and practical frameworks. Key questions remain regarding how GenAI, especially ChatGPT, affects second language

acquisition, its mechanisms, sustainability, and associated risks. To address these gaps, this study conducted a scoping literature review using ChatGPT as a representative GenAI tool, following the three stages of planning, execution, and reporting under PRISMA 2020 guidelines. The process included dual-reviewer screening with explicit inclusion and exclusion criteria, transparent documentation via a PRISMA flow diagram, and rigorous quality assessments to reduce bias. Systematic searches across Google Scholar, Scopus, and Dialnet initially identified 1123 records. After duplicates and title/abstract screening, 74 articles proceeded to full-text assessment, and 35 studies published between 2023 and 2025 were selected for in-depth analysis. The review highlights four key dimensions through which GenAI supports second language teaching and learning: personalization, autonomy, immediate feedback, and critical thinking. These were documented across qualitative (14%), quantitative (23%), mixed-methods (11%), and review studies (40%) and are considered essential for enhancing instructional quality. Additionally, the study identifies methodological shortcomings, contradictions, and research gaps such as limited attention to non-ChatGPT tools, scarce empirical classroom data, and language bias, while also offering targeted recommendations and strategies for future research and pedagogical practice, thereby advancing theoretical understanding and improving teaching effectiveness.

Keywords: Generative artificial intelligence (GenAI). ChatGPT. Scoping literature review (SLR). Second language learning.

Resumen

En los últimos años, IA ha avanzado rápidamente, mostrando un notable potencial en la enseñanza y aprendizaje de lenguas extranjeras. Sin embargo, como tecnología emergente, su integración educativa sigue siendo exploratoria. La mayoría de los estudios se centran en funcionalidades aisladas o hallazgos limitados, careciendo de marcos teóricos y prácticos sólidos. Persisten preguntas sobre cómo

la GenAI, especialmente ChatGPT, afecta la adquisición de segundas lenguas, sus mecanismos, sostenibilidad y riesgos asociados. Para abordar estas lagunas, se realizó una revisión exploratoria de la literatura usando ChatGPT como herramienta representativa de GenAI, siguiendo las etapas de planificación, ejecución e informe según PRISMA 2020. El proceso incluyó revisión doble con criterios explícitos de inclusión y exclusión, documentación transparente mediante un diagrama PRISMA y evaluaciones rigurosas de calidad para minimizar sesgos. Las búsquedas sistemáticas en Google Scholar, Scopus y Dialnet identificaron 1123 registros. Tras eliminar duplicados y revisar títulos y resúmenes, 74 artículos pasaron a evaluación de texto completo, y 35 estudios publicados entre 2023 y 2025 fueron analizados en profundidad. La revisión identifica cuatro dimensiones clave mediante las cuales GenAI respalda la enseñanza de segundas lenguas: personalización, autonomía, retroalimentación inmediata y pensamiento crítico. Estas se documentan en estudios cualitativos (14 %), cuantitativos (23 %), de métodos mixtos (11 %) y revisiones (40 %), y son esenciales para mejorar la calidad instruccional. Además, se señalan limitaciones metodológicas, contradicciones y vacíos, como poca atención a herramientas distintas de ChatGPT, escasez de datos empíricos y sesgo lingüístico, ofreciendo recomendaciones para investigación futura y práctica pedagógica.

Palabras clave: Inteligencia artificial generativa (GenAI). ChatGPT. Revisión exploratoria de la bibliografía (SLR). Aprendizaje de segundas lenguas.

1. INTRODUCTION

John McCarthy coined the term "artificial intelligence" (AI) in 1956. Over the ensuing decades, AI research has passed several important milestones. In the 1970s and 1980s, the emergence of expert systems and their probabilistic counterparts marked the first major advances in rule-based reasoning and uncertainty handling.

Later, the introduction of backpropagation revitalized artificial neural networks (Wexell Machado & Canese, 2024).

With OpenAI's release of ChatGPT in November 2022, adoption accelerated dramatically, the service attracted more than one million users within its first five days and has since been applied in fields such as economics and finance, journalism, medicine, and education (García-Méndez et al., 2025; Monib et al., 2024). From GPT-3 and GPT-3.5 through GPT-4 and GPT-4.5 to the more recent GPT-5 series—including models such as GPT-5.3 and GPT-5.4—GenAI has evolved rapidly. The swift, iterative development of GenAI and its rapid integration into education have outpaced the long validation cycles typical of new instructional materials. ChatGPT, developed by OpenAI, is a conversational agent underpinned by large language models (LLMs) which enable it to interact with users in natural, human-like dialogues and provide contextually relevant educational support (Monib et al., 2024; Xiao & Zhi, 2023; Karataş et al., 2024; Lo et al., 2024). Advances exemplified by ChatGPT show that GenAI can produce outputs that closely resemble human-authored content, it accepts multimodal inputs, leverages vast parameter counts reportedly on the order of one hundred trillion parameters and, through training on extensive text corpora, captures linguistic nuance to generate contextually appropriate responses. GenAI tools simplify aspects of scholarly writing by providing direct answers and reducing the need for manual literature searches. They have proven effective for tasks such as article generation, complex question answering, pronoun-noun agreement, sentiment analysis, summarization, code generation, and storytelling (Law, 2024).

AI-driven language-learning tools have consequently attracted intense interest from researchers and practitioners. GenAI has captured global attention because LLMs such as the GenAI family are capable of producing coherent, grammatically correct text and adapting to user interactions. The advent of GenAI most notably ChatGPT has far-reaching implications for many fields, including language pedagogy. GenAI systems can generate textual, visual, and other digital materials that students can engage with; their applications range from speech recognition and conversational

assistants to information retrieval (Chiu & Sanusi, 2024). ChatGPT can answer questions, draft texts, perform logical reasoning, debug code, translate languages, and has already demonstrated undeniable pedagogical value in educational contexts (Manzanedo, 2025; Wei, 2023).

Language teaching remains a cornerstone of global education, in an increasingly interconnected world, effective communication and intercultural understanding are essential for academic, professional, and personal success. Mastery of a second language enables meaningful interaction, clear expression of ideas, and navigation of diverse cultural contexts. Traditionally, language educators have served as the principal facilitators of language acquisition within formal settings, assessing student progress through written and oral evaluations. However, the rise of the Internet and search engines has reshaped the learning landscape by substantially reducing learners' dependence on instructors. Students now have access to vast information repositories, language resources, and learning platforms that cater to individualized needs. The introduction of GenAI projects represents a further paradigm shift in language education and acquisition (Law, 2024).

Integrating GenAI into language teaching can not only enhance linguistic competence but also boost learner motivation and engagement, offering an innovative pedagogical strategy for addressing twenty-first-century educational challenges (Changoluisa Santacruz et al., 2024). The deployment of GenAI in language classrooms is therefore both an important step toward optimizing instructional effectiveness and a key avenue for renewing pedagogical practice (Wexell Machado & Canese, 2024). In the current era of classroom digitization, GenAI has emerged as a transformative tool that is redefining approaches to language teaching and learning (Gibert et al., 2024).

1.1. Rationale, Objectives and Research Questions

Despite the expanding application of GenAI across sectors including education important gaps remain. While prior studies have documented the potential value of GenAI for language learning (Chicaíza et al., 2023; Marzano, 2025; Mah et al., 2025), Mah et al. (2025) note a shortage of forward-looking analyses that examine the prospective impacts of GenAI on instructional practice.

Existing literature reviews suggest that tools such as ChatGPT can support aspects of foreign language learning, yet most studies address only isolated functionalities or short-term interventions. This raises a central question for the field:

RQ: In which concrete dimensions can GenAI most substantially support the teaching and learning of a second language?

To fill this gap, this study conducts a scoping review guided by PRISMA principles and complementary meta-analytic techniques. By mapping and synthesizing empirical and theoretical work published between 2023 and 2025, the review aims to

(a). Delineate the concrete pedagogical, curricular, assessment, and affective/cognitive dimensions where GenAI is being applied.

(b). Evaluate the strength and consistency of evidence across those dimensions.

(c). Identify key research gaps and practical implications for educators, policymakers, developers, and researchers. The resulting synthesis will clarify current trends, highlight methodological limitations in the emerging literature, and propose prioritized directions for future research and practice (Lo et al., 2024; Law, 2024).

2. THEORETICAL FRAMEWORK

We reviewed prior research and confirmed the significant role of artificial intelligence in second-language instruction. In particular, GenAI systems such as ChatGPT support four interrelated dimensions personalization, learner autonomy, immediate corrective feedback, and critical thinking, that are central to second-language acquisition theory. These dimensions not only offer concrete support to learners during the acquisition process but also provide a sound theoretical and practical basis for pedagogical innovation and the enhancement of language teaching.

2.1. The contribution of GenAI to second language teaching

The recent technical maturation of GenAI has created new, practical opportunities for second-language (L2) pedagogy. Advances in reinforcement learning from human feedback and alignment techniques illustrated by systems such as DeepSeek-R1 have demonstrably improved multi-step reasoning and the instructional alignment of model outputs, making GenAI tools more capable of producing coherent, pedagogically oriented explanations and scaffolded interactions that support learner understanding and progression (Guo et al., 2025). These advances enable models to generate stepwise hints, scaffolded corrections, and progressive prompts that can be directly embedded within language tasks to support incremental skill development.

Large multimodal foundation models now offer integrated text, audio and visual generation and interpretation, enabling richly contextualized L2 practice. Multimodal capabilities support tasks such as image-prompted speaking tasks, video captioning and discussion, synthetic conversation partners with aligned visual context, and audio example generation for pronunciation practice each expanding authentic practice opportunities and increasing learner engagement through varied input channels (Küchemann et al., 2025; Zhang et al., 2024). The ability to produce tailored multimodal materials rapidly allows teachers to design topic-relevant, culturally

situated activities at scale, which can better match learners’ interests and curricular goals.

A growing and diverse empirical literature documents measurable gains associated with pedagogically integrated GenAI tools. Meta-analytic evidence indicates positive effects of ChatGPT-style interventions on student learning performance, perceived learning, and higher-order thinking, with consistent reports of improved engagement and problem-solving dispositions when AI tools are used as part of structured learning activities (Wang & Fan, 2025). Large randomized and quasi-experimental studies further demonstrate that AI-driven tutoring systems and adaptive practice regimes can produce substantial improvements on targeted learning outcomes, in some cases outperforming conventional active-learning formats for particular skills and domains (Kestin et al., 2025; Fan et al., 2025).

In L2-specific applications, GenAI enhances several instructional levers that directly support language development. First, LLMs efficiently generate iterative, item-level feedback on learners’ written and spoken productions providing alternative phrasing, targeted corrective sequences, and exemplars that increase practice volume and immediacy of feedback available to each learner (Er et al., 2024; Banihashem et al., 2024). Second, adaptive sequencing and task modification driven by modelled learner profiles allow bespoke practice schedules and scaffolded challenge adjustments that align with individual proficiency trajectories and curricular standards (Kikalishvili, 2024; Kestin et al., 2025). Third, teachers can generate tailored dialogues, simulated interlocutors, role-play scenarios, audio samples, and visual prompts that mirror learners’ topical interests and proficiency levels vastly reducing preparation time while increasing the relevance and authenticity of practice (Hemminki-Reijonen et al., 2025; Küchemann et al., 2025). Fourth, several evaluations report promising concordance between model-generated assessments and criterion-based grading rubrics, indicating potential for GenAI to support scalable, consistent formative assessment workflows and to assist in rubric-aligned scoring practices (Zhang et al., 2024).

Synergies between GenAI and immersive or situated learning environments produce particularly compelling instructional innovations. Integrations of GenAI with virtual reality (VR) and experiential platforms enable situated communicative practice with contextually grounded feedback, support reflective learning loops, and create realistic interlocutor simulations that deepen transfer to real-world communicative tasks (Hemminki-Reijonen et al., 2025; Wang et al., 2024). When used to generate scenario-specific prompts, role-plays and post-task reflective questions, GenAI enhances the authenticity of practice and supports the development of pragmatic and interactional competencies.

Practically, a range of applied studies and implementation reports provide concrete guidance for leveraging GenAI to enhance L2 teaching. Prompt pattern catalogs and implementation guides furnish teachers with reusable templates for eliciting pedagogically rich responses, thereby streamlining prompt engineering and enabling consistent instructional use across different classroom contexts (White et al., 2023). Case studies and empirical reports from STEM and humanities education illustrate transferable workflows such as integrating AI-generated exemplars into lesson sequences, embedding iterative feedback loops into writing tasks, and deploying adaptive practice modules for differentiated instruction that have produced observable gains in learner outcomes and engagement (Wang et al., 2023; Qadir, 2023; Ellis & Slade, 2023; Mohammed et al., 2025).

In sum, converging technical advances, multimodal affordances, meta-analytic support, and domain-specific implementations indicate that GenAI can substantially enrich L2 pedagogy by expanding the scale and quality of formative feedback, enabling personalized and multimodal practice, supporting immersive authentic tasks, and streamlining materials creation and assessment processes. These contributions position GenAI as a powerful complement to existing instructional approaches, providing practical pathways for increasing individualized practice, diversifying input, and accelerating targeted skill development in second-language classrooms.

2.2. Theories

2.2.1. Personalization

A key advantage of GenAI in language learning is its ability to provide highly personalized experiences tailored to each learner's needs (Sasmayunita & Assiddiq, 2025). Personalization in second language instruction is not just about grouping learners by proficiency level; it involves offering support that enables each learner to progress at their own pace. When teachers adapt content to learners' differences and provide customized feedback based on ongoing performance, learners' motivation and confidence typically increase significantly. Formative assessment plays a central role in this process, as it identifies misunderstandings or difficulties promptly, allowing for adjustments in instructional strategies. Because each learner follows a unique learning trajectory, feedback must also be individualized, providing specific, actionable guidance based on the learner's current knowledge state rather than generic correctness judgments. Effective personalized feedback identifies concrete obstacles and proposes targeted solutions; each feedback session should reflect individual needs rather than standardized benchmarks (William, 2011).

The importance of personalization in foreign-language education is increasingly recognized. From the perspective of the diffusion of innovations, introducing personalized learning in language instruction represents an innovation whose relative advantage lies in its ability to tailor content to individual needs. Personalization, delivered through customized learning pathways and focused support, can enhance learner engagement, motivation, and outcomes (Sasmayunita & Assiddiq, 2025; Xue, 2025; Wu, 2024). Tomlinson (2001) further argues that content personalization can be achieved in two complementary ways: modifying the content itself and varying how learners encounter it. Generally, maintaining core content while adjusting its presentation to learners' readiness, interests, or learning preferences is most effective.

2.2.2. *Autonomy*

Learner autonomy is a critical construct in second language education. Autonomy refers to a learner's readiness to take responsibility for their own learning and includes three dimensions: self-management, self-regulation, and self-determination (Dickinson, 1995). Autonomy activates intrinsic motivation by satisfying basic psychological needs: competence is enhanced through task mastery and self-efficacy, relatedness is fostered when choice facilitates authentic collaboration, and autonomy is reinforced through freedom of choice. On the other hand, externally imposed structures can undermine motivation, leading to resistance or passive compliance. Providing personalized, independent learning environments enhances learners' autonomy and enthusiasm, increasing motivation. Learners who can practice a language anytime and anywhere gain greater control over their learning, fostering both autonomy and self-regulation (Wei, 2023).

Dickinson (1995) emphasizes that autonomy is not a teaching technique but a learner capability, manifested in independence from routine instruction and enacted through decisions about goals, content, methods, monitoring, and assessment. Technological tools have redefined autonomy by enabling independent access to language resources. However, genuine autonomy requires algorithmic literacy, allowing learners to critically engage with digital scaffolds rather than passively accepting their outputs. Autonomy presupposes interdependence, not isolation; learners should be prepared to accept responsibility for outcomes when collaborating with teachers and peers on goal setting. In practice, autonomy is constructed through micro-decisions, such as selecting word lists or peer reviewers, which support macro-level autonomy, like designing long-term learning paths. Crucially, autonomy is acquired, not simply taught. While educators can create enabling conditions, self-management behaviors remain the learner's responsibility.

In language learning, learner autonomy is increasingly viewed as a key determinant of success. Learner autonomy refers to learners' ability to independently manage their learning processes setting goals, selecting strategies, and evaluating

outcomes and has become central to contemporary pedagogy, especially in language education, where learners are encouraged to take greater responsibility for their learning journeys (Sasmayunita & Assiddiq, 2025).

2.2.3. Immediate feedback

Immediate feedback helps learners identify and correct errors in real time, improve language production, and develop self-monitoring and self-correction skills (Qiao & Zhao, 2023). Timely feedback fosters autonomy and responsibility by eliminating the need for teacher corrections. By making interactions more immediate and engaging, instant feedback reduces the cognitive load of delayed corrections, allowing learners to apply corrections to subsequent attempts without revisiting previous task contexts. In language learning, especially oral communication that demands prompt responses, direct interaction remains indispensable (Sasmayunita & Assiddiq, 2025).

As constructivist theory emphasizes, feedback acts as a scaffold, guiding learners to higher understanding levels (Xue, 2025). Immediate feedback is central to best practices in second language instruction. It is valuable only when it influences learners' subsequent performance on similar tasks. Thus, timeliness is not just about speed; feedback must be delivered while learners remain focused on the learning goal and task context, with an opportunity to act on it (Van den Bergh et al., 2006). Delayed feedback has limited value; feedback delivered near task completion is most impactful, as the learner's understanding of goals, strategies, and context remains fresh. Immediate feedback links actions to responses, encouraging deeper reflection and adjustments. Effective feedback should be provided within the learning cycle so learners can use it to refine ongoing or upcoming tasks. Equally important is the quality of feedback its clarity, specificity, constructiveness, and alignment with key learning objectives. Rushed or superficial feedback can be harmful. Feedback timed to support specific learning processes is more valuable than absolute immediacy (Shute, 2008). Teachers can achieve meaningful immediacy by leveraging AI technologies, designing

assessments that facilitate efficient feedback, or staging feedback delivery (Nicol & Macfarlane-Dick, 2006).

Bitchener & Knoch (2008) emphasize that timely feedback should be provided close to learner output to correct errors, prevent fossilization, reinforce correct responses, and maintain engagement. In well-designed environments, such as AI tutoring systems, immediate feedback serves as scaffolding during skill acquisition. In dynamic learning contexts, it guides exploration, reduces frustration, and, by lowering cognitive load, frees up resources for deeper learning. Immediate feedback is most effective for novices, tasks involving procedural knowledge or motor skills, and for maintaining motivation in challenging situations. As proficiency increases, feedback can be gradually withdrawn to foster self-assessment.

2.2.4. *Critical thinking*

Critical thinking has become a key educational objective globally due to its contribution to lifelong learning in recent decades (Liang & Wu, 2024). Fostering critical thinking has become a central aim in education, beyond mere knowledge and skill acquisition (Yatani et al., 2024). Critical thinking facilitates the interplay between linguistic form and content construction; without it, learners struggle to develop independent ideas. There is a symbiotic relationship between critical thinking and argumentative writing, which naturally supports second language learners' linguistic development. Critical and creative thinking are skills that can be developed through instruction and practice in reasoning, analysis, planning, and questioning (Suh et al., 2025). Moreover, the attitudinal components of critical thinkers such as the willingness to plan, cognitive flexibility, perseverance, readiness to self-correct, sustained focus, and the pursuit of consensus are central to this construct (Yatani et al., 2024).

In the context of second language instruction, critical thinking is defined as purposive, self-regulated judgment involving interpretation, analysis, evaluation, inference, and articulation of the considerations underlying one's judgments, whether evidential, conceptual, methodological, or contextual. It involves applying cognitive

strategies to increase the likelihood of achieving desired outcomes and entails goal-directed, rational reasoning for problem solving, inference construction, and decision making while minimizing cognitive biases. Critical thinking requires reflection on beliefs and actions, often entailing discomfort, as it demands acknowledgment of one's ignorance and a willingness to challenge dogma (Facione, 2011). In the face of the rapidly changing educational landscape, critical thinking is indispensable (Sasmayunita & Assiddiq, 2025).

Critical thinking is the deliberate application of cognitive skills or strategies to enhance the probability of desirable outcomes. It is purposeful, rational, and goal-directed, encompassing problem solving, inference, probabilistic reasoning, and decision making. These skills involve metacognition reflection on one's own thinking and the willingness to engage in effortful thought and revise beliefs when warranted (Halpern, 2014). Identifying assumptions is core to critical thinking, and this process varies with pedagogical context and intellectual traditions. Key steps in learning include uncovering and testing assumptions, exploring alternative viewpoints, and taking informed action. Assumptions are rarely absolutely true or false but are contextually situated (Brookfield, 2013). Elder & Paul (2007) emphasize that critical thinking requires self-direction, discipline, monitoring, correction, adherence to rigorous standards, effective communication, and a commitment to overcoming egocentric and sociocentric biases.

In sum, the efficacy of second language teaching depends substantially on integrating relevance, personalization, autonomy, immediate feedback, and critical thinking. These interrelated dimensions together form a dynamic, interactive, and effective learning ecosystem. In the context of rapid educational-technology development, leveraging these elements shifts instruction from mere knowledge transmission to deeper cognitive growth, enabling second language learners to acquire genuine linguistic competence alongside affective and social-interactional capacities.

3. METHODS

This study adopts the concept of a “scoping review”, which was first proposed by Arksey & O'Malley (2005), who established the initial methodological framework for scoping reviews. Subsequently, Levac et al. (2010) refined this framework, emphasizing more systematic steps and greater transparency. Later, Peters et al. (2015) also issued relevant guidelines, gradually standardizing the approach. A scoping literature review is a structured approach to evidence synthesis that is governed by a predefined review protocol setting out each step of the process in advance. Consistent with best-practice reporting standards, the scoping literature review will follow the PRISMA 2020 guidelines. Study selection will be documented using a PRISMA flow diagram and checklist items, and a dual-reviewer screening procedure with explicit inclusion and exclusion criteria will be implemented. In addition, quality assessments of included studies will be conducted to mitigate the risk of bias (Lo et al., 2024; Ogunleye et al., 2024).

Once the research questions have been specified, a comprehensive search strategy will be developed to cover multiple bibliographic databases and grey literature sources, thereby maximizing the likelihood of capturing all relevant evidence. Reviewers will independently screen records and extract data according to the pre-established protocol; this independence both reduces bias and allows for the deliberate inclusion of discrepant or contradictory findings. Each primary study will be appraised using established quality-assessment instruments. Where sufficient quantitative data exist, meta-analysis will be performed to integrate results, identify patterns of robustness or sources of heterogeneity across contexts, and detect effects that single small studies might overlook.

Although the scoping literature review approach demands substantial time and resources, it yields highly reliable evidence that can meaningfully inform practice and future research. Methodologically, the framework comprises three stages: planning, conduct, and reporting and culminates in a transparent final report that details the

search strategy, presents applicability and quality assessments, and documents the methodological rigor underpinning the review’s conclusions.

3.1. Planning stage

In this stage, the review protocol is developed with the goal of minimizing bias and ensuring transparency throughout the process (Changoluisa Santacruz et al., 2024). The protocol includes the following components:

1. Research questions

Formulating precise questions is essential to guide the search and selection of sources. The central question of this study is: In which specific dimensions can GenAI most significantly support the teaching and learning of a second language?

2. Search strategy

To retrieve relevant literature, three academic repositories were consulted: Google Scholar, Scopus and Dialnet. Search queries were conducted using combinations of key terms in both English and Spanish, as well as their synonyms. The main search strings used were:

C1: “AI ChatGPT as a second foreign language” or “AI second foreign language”

C2: Spanish equivalents.

Table 1 provides a detailed account of the number of records retrieved from each database using these search queries.

Database	C1	C2	Total
Google Scholar	154	216	370
Scopus	417	0	417
Dialnet	332	4	336
Total	903	220	1123

Table 1. The articles obtained from scientific databases

3.2. Primary study selection procedure

In the initial phase, all documents whose titles and abstracts explicitly align with the study's scope were compiled in a spreadsheet. Duplicate records, resulting from their appearance across multiple databases, were immediately removed. The refined list then underwent a second round of evaluation based on the strict application of the inclusion and exclusion criteria described in Table 2. This screening process considered factors such as publication period, type of media, language, document category, and source of origin ensuring that only those studies meeting the established criteria remained for analysis. Once this processing was completed, the selected articles constituted the final corpus of primary studies that would inform the scoping literature review, ensuring a coherent approach aligned with the research objectives.

Category	Inclusion criteria	Exclusion criteria
Research type	Detailed descriptions of AI-involved teaching processes	Articles that are not accessible in full text
Research subjects	Formal educational settings	Informal learning environments (e.g., self-study at home, non-accredited institutions)
Language	Written in English or Spanish	Written in other languages
Intervention	AI tools like ChatGPT	Non-AI technological interventions (e.g., traditional multimedia teaching)
Outcome measures	Focused on improving personalization in learning	No specific teaching outcomes or irrelevant to the research objectives
Time range	Published from January 2023 to June 2025	Published before January 2023

Table 2. Inclusion and exclusion criteria

In the third stage of review, these articles were assessed for their relevance to the study topic using the relevance criteria shown in Table 3. Each article was rated on a Likert scale based on its degree of relevance. Finally, the relevant information from the selected studies was recorded in a summary matrix.

Relevance criteria
Personalization
Autonomy
Immediate feedback
Critical thinking

Table 3. Relevance criteria

During the first screening, I identified 1123 preliminary records relevant to my research questions. In the second screening, I applied the predefined inclusion and exclusion criteria, resulting in 74 articles eligible for further consideration.

During the third screening, each of these 74 articles was evaluated using a five-point Likert scale based on the specified relevance dimensions. The response options were: a) Strongly disagree; b) Disagree; c) Neutral; d) Agree; e) Strongly agree.

Each relevance criterion was assigned a score ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). These scores were then grouped into two aggregates:

$$\Sigma(a + b)$$

$$\Sigma(c + d + e)$$

An article was retained for in-depth analysis if $\Sigma(c + d + e)$ exceeded $\Sigma(a + b)$, as illustrated in Table 4. Articles that met the basic eligibility requirements but for which $\Sigma(a + b) \geq \Sigma(c + d + e)$ were noted as eligible but were not advanced to full analysis (Changoluisa Santacruz et al., 2024).

Relevance criteria	a	b	c	d	e
Personalization					1
Autonomy		1			
Immediate feedback			1		
Critical thinking	1				1
Total		2		3	

Table 4. Example of evaluation of articles using the Likert scale

As a result of the evaluation, 35 documents that meet the established criteria and are relevant to the research were selected Fig. 1.

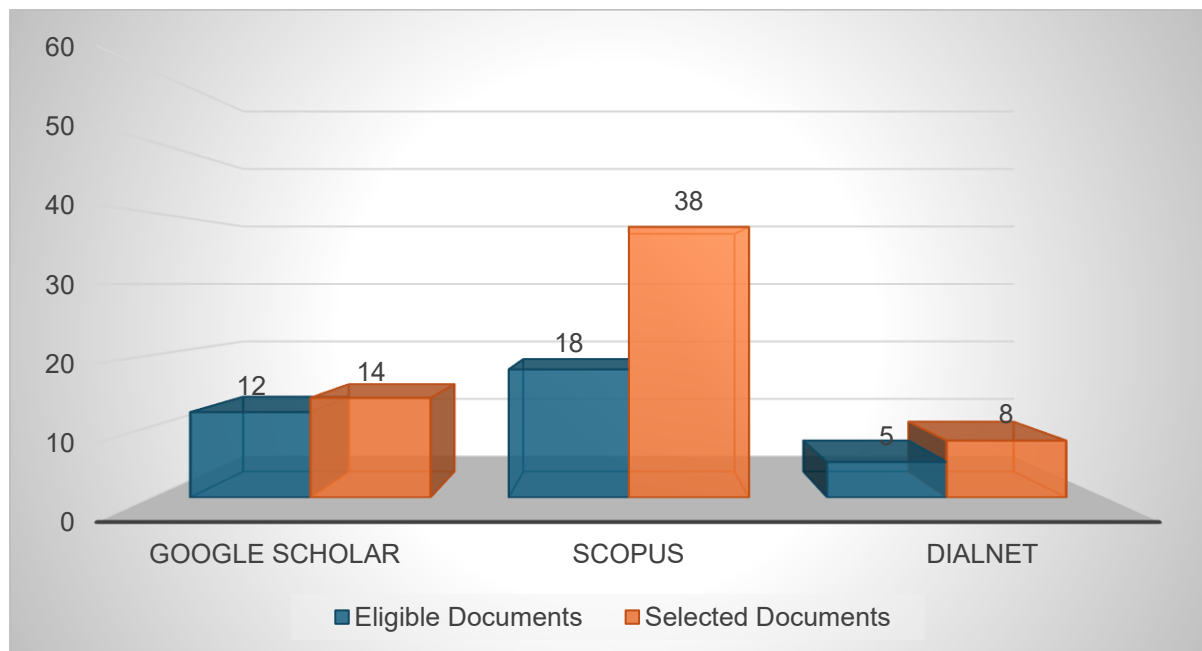


Fig. 1. Articles analyzed for scoping literature review

Next, Fig. 2 illustrates the vertical hierarchical structure of the scoping review conducted in this study. At the top level, “Scoping Review” represents the overall methodological framework for evidence synthesis. The second level depicts the process of screening and selecting primary studies, including the total number of records initially identified (1123), the removal of duplicates, and the application of inclusion and exclusion criteria, resulting in 74 articles eligible for further evaluation. The third level presents the relevance assessment of the selected studies, structured according to four key dimensions: personalization, autonomy, immediate feedback, and critical thinking. Finally, the fourth level summarizes the outcome of the assessment, showing that 35 studies met all criteria and were included in the in-depth analysis. This vertical layout clearly conveys the logical progression from general methodology to specific evaluation criteria, offering an intuitive and comprehensive visual summary of the study’s literature handling, screening, and assessment framework.

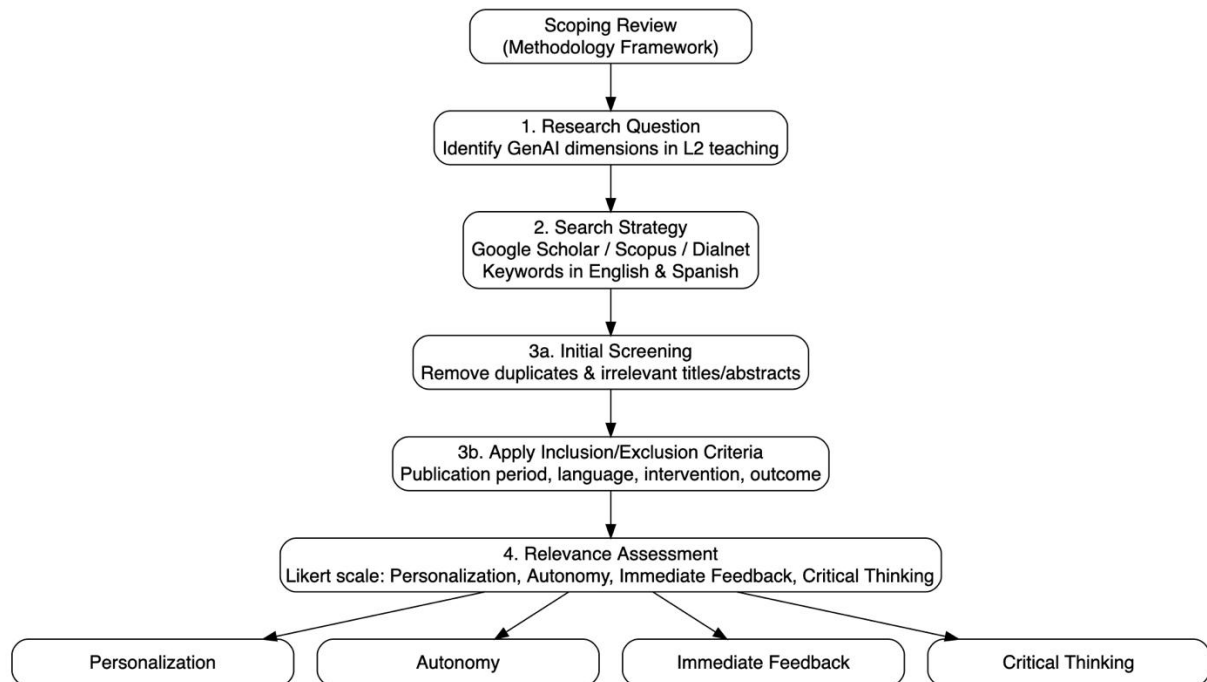


Fig. 2. Hierarchical structure of the scoping literature review and relevance assessment

4. RESULTS

4.1. Selection of sources of evidence

During the identification phase, searches in Google Scholar, Scopus, and Dialnet yielded 1123 records, with no additional records from other sources. All citation data were exported to a spreadsheet for processing. After removing duplicates and screening titles and abstracts, 74 records (6.59% of the initial yield) advanced to the full-text assessment stage. Each of these 74 articles was fully examined and assessed against the predefined inclusion and exclusion criteria. At this stage, 39 articles were excluded (52.70% of the full-text assessment sample; 3.47% of the initial search yield). A total of 35 studies met all inclusion criteria and proceeded to full-text coding and synthesis (3.12% of the initial search yield; 47.30% of those assessed in full text). The

screening process was conducted independently by two reviewers, with disagreements resolved through discussion. The number of records and reasons for exclusion at each step are reported according to the PRISMA 2020 flow diagram (see Fig.2).

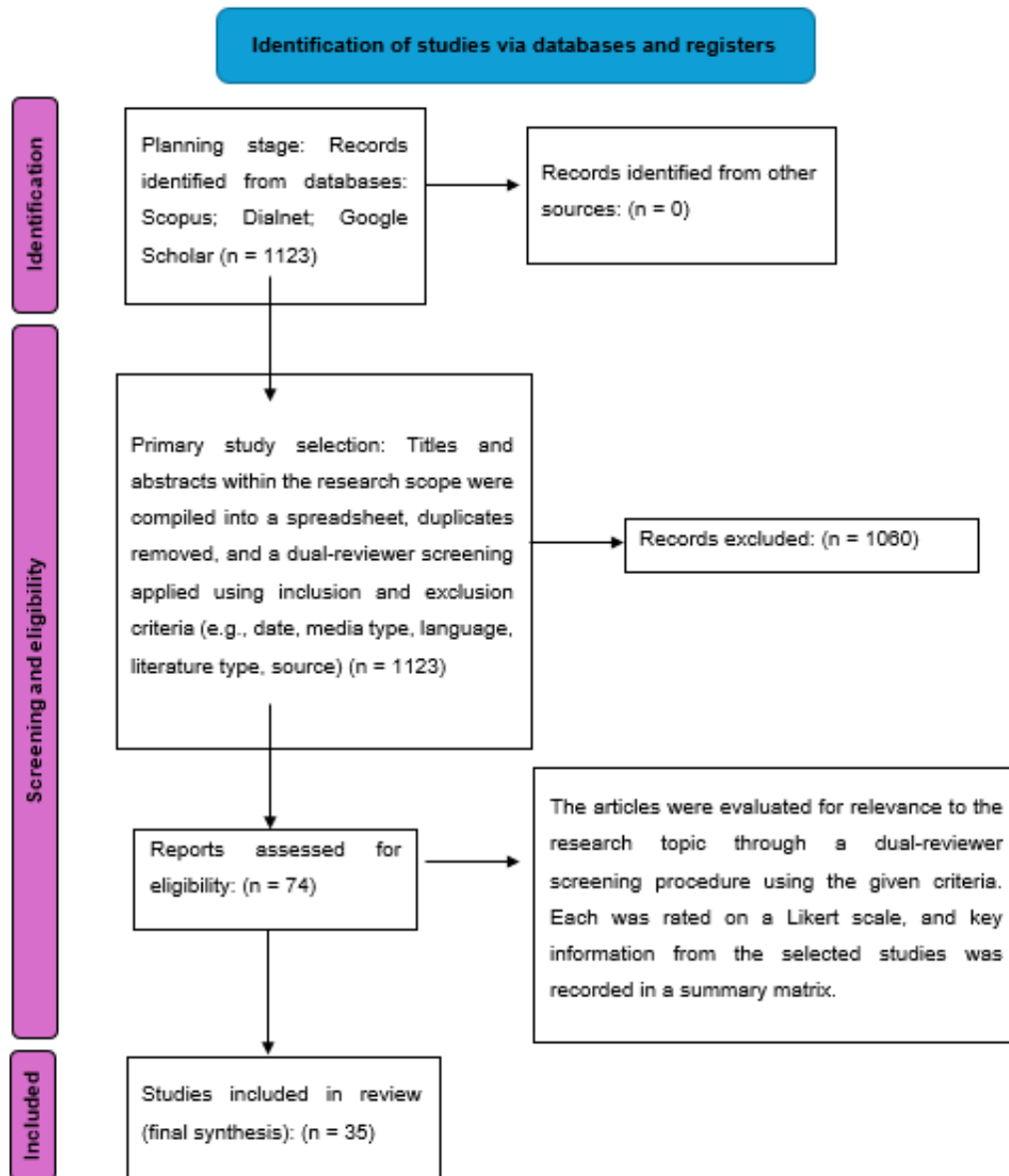


Fig. 3 PRISMA 2020 flow diagram

4.2. Characteristics of sources of evidence

The analysis of 35 studies published between 2023 and 2025 (100% of the sample) reveals significant methodological diversity, with a strong focus on GenAI's role in enhancing second language learning across four key dimensions: personalization, autonomy, immediate feedback, and critical thinking.

Five studies (14%) used qualitative approaches, mainly case studies and exploratory designs, to examine how GenAI fosters learner autonomy, supports personalized pathways, provides immediate feedback, and stimulates critical thinking. Four studies (11%) employed mixed methods, combining quantitative and qualitative data to assess GenAI's impact on personalized instruction, learner self-regulation, and critical cognitive skills development. Eight studies (23%) were quantitative, including surveys, experimental designs, pre-test and post-test studies, measuring GenAI-assisted interventions' effectiveness in delivering personalized learning, enhancing autonomous learning, and providing real-time corrective immediate feedback.

Literature and systematic reviews made up 14 studies (40%), summarizing evidence on GenAI's contribution to personalization, immediate feedback, autonomy, and critical thinking. Three studies (9%) were technical reviews or exploratory investigations, focusing on the practical application of GenAI tools to optimize these four dimensions in the classroom. One studies (3%) were commentaries or editorial perspectives, offering insights into GenAI's potential to strengthen these aspects of second language education.

Overall, the methodological diversity highlights the various ways GenAI supports key aspects of effective foreign-language instruction. Detailed information is presented in Table 5.

Research method	Number of studies	Percentage (%)
Qualitative research	5	14
Mixed-methods research	4	11
Quantitative research	8	23
Literature review / Systematic review	14	40
Technical review / Exploratory study	3	9
Commentary / Editorial position	1	3
Total	35	100

Table 5. Methodological distribution of 35 chosen studies published between 2023 and 2025 on
GenAI in second language teaching

A comprehensive heatmap (Fig. 3) illustrates how the 35 selected studies address the four key dimensions personalization, autonomy, immediate feedback, and critical thinking using a 5-point Likert scale.

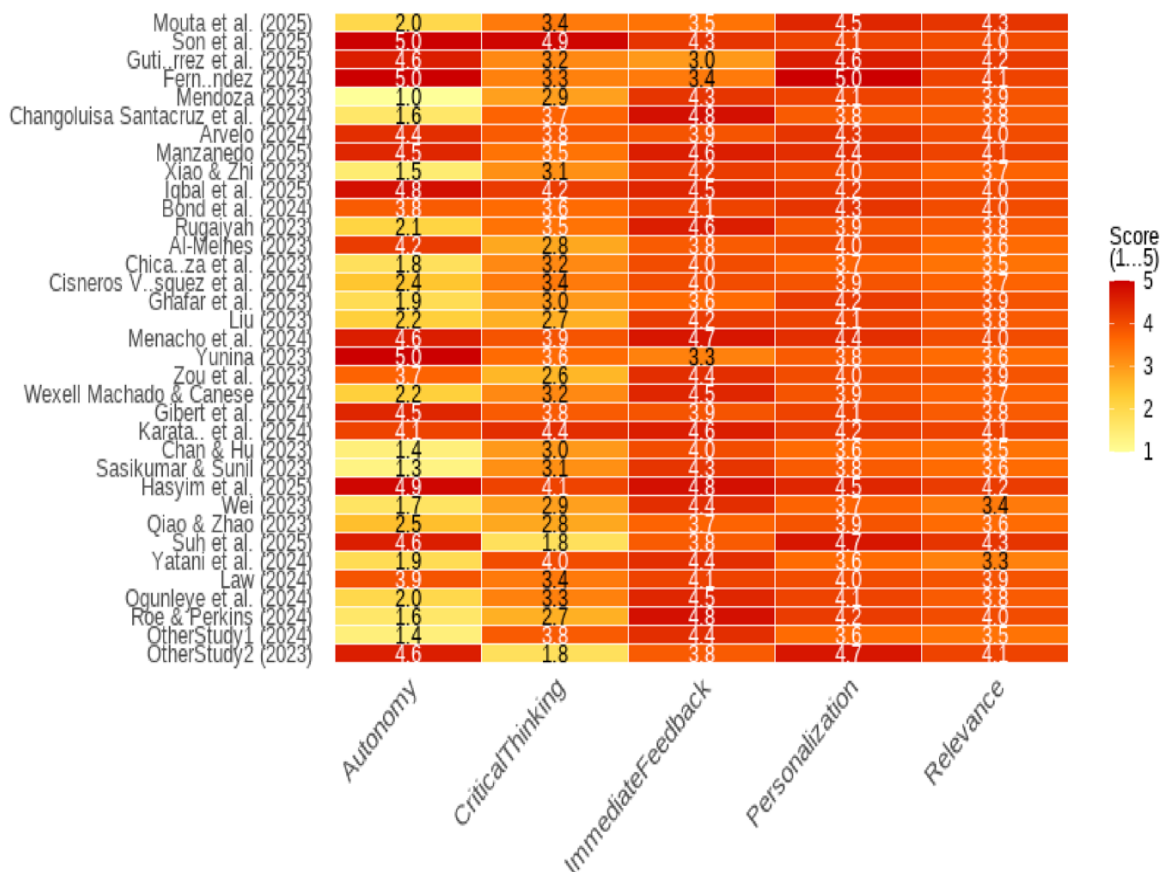


Fig. 3. Heatmap of Dimensional Support by GenAI in 35 Second-Language Studies

The heatmap reveals several notable patterns. Personalization and immediate feedback consistently receive high scores across the majority of studies, indicating that these aspects are both widely emphasized in research and robustly implemented in instructional practice. Autonomy and critical thinking also receive relatively high scores overall, with most studies demonstrating strong alignment between course content or tasks and intended learning objectives.

Autonomy, in contrast, exhibits pronounced polarization: while some studies provide exceptionally high support for learner autonomy, others show minimal attention to fostering independent learning. Critical thinking demonstrates the greatest variability; certain studies explicitly prioritize the cultivation of higher-order cognitive skills, whereas many others score low in this dimension, reflecting an uneven emphasis on critical thinking in both research and practice.

The heatmap also highlights a few studies with relatively balanced and comprehensive profiles. For example, some investigations achieve high scores in personalization, immediate feedback, and autonomy, yet often display divergence in autonomy or critical thinking, with certain studies excelling in the former dimensions but underperforming in the latter. Overall, instructional interventions tend to concentrate on enhancing immediate corrective feedback and personalized support, whereas integrative approaches that simultaneously foster both autonomy and critical thinking remain scarce, indicating a need for further research and practical validation in this area.

Next, the study addresses the following research question by analyzing the findings from the scoping literature review on the 35 studies that met the final inclusion criteria: In which specific dimensions does GenAI most effectively support second-language teaching?

Mouta et al. (2025) demonstrate that a systematic literature review is an effective method for exploring language learning personalization. Their study shows that GenAI can design personalized learning pathways, allowing students to adjust the pace according to their needs and thereby improving outcomes. These advantages

are particularly evident in enhancing personalized teaching, fostering self-directed learning, and promoting critical thinking skills, which are vital for long-term academic success.

Son et al. (2025) further show that GenAI tools provide students with personalized, interactive guidance. In language teaching, GenAI enables individualized learning without direct teacher intervention, offering flexible environments that combine immediate feedback and learner autonomy, thus helping students face the challenges of foreign language acquisition more effectively.

Fernández (2024), in a literature review, emphasizes that integrating GenAI into language instruction through personalization and immediate feedback makes it a highly effective teaching resource. This approach advances personalized teaching, fosters students' autonomy, and significantly enhances their technological proficiency.

Mendoza (2023), in a qualitative exploratory case study, reports that deploying ChatGPT for out-of-class learning focused on autonomy helps personalize adjustments to meet diverse student profiles. This approach promotes autonomous learning strategies and helps second language learners improve proficiency independently.

Gutiérrez et al. (2025) highlight ChatGPT's potential in foreign language teaching, particularly for advancing personalization and critical thinking, enabling learners to better manage the development of their grammatical skills and autonomy in the target language.

Changoluisa Santacruz et al. (2024), in a systematic literature review, analyzed various AI tools used in language learning and found that these systems can provide genuinely personalized feedback. Tools like ChatGPT, with their advanced personalization algorithms, have demonstrated the ability to deliver tailored feedback to each learner, thereby enhancing autonomy and optimizing the language learning environment.

Arvelo (2024), via a systematic qualitative analysis of available AI tools, reveals that although this learning model may seem informal, it enables students to take an autonomous leading role outside the classroom, creating a personal learning environment (PLE) that emphasizes personalization. This approach fosters critical thinking and independent learning capabilities while extending the reach of language instruction across varied contexts. ChatGPT stands out for its ability to generate comprehension questions, design personalized teaching materials, and provide immediate feedback to both teachers and students, driving the development of critical skills and deep self-directed learning.

Manzanedo (2025), using a mixed-methods approach that combined an SLR and a quantitative survey, demonstrates that GenAI can continuously provide personalized conversational practice, with immediate feedback as the key element of adaptive learning. This capability allows instructional activities to be adjusted in real time according to each student's progress, thereby improving educational outcomes. As a chatbot capable of delivering immediate feedback, ChatGPT offers students essential opportunities for live conversational practice, facilitating direct interaction that improves both the quality of oral exercises and the effectiveness of feedback.

Xiao & Zhi (2023), in a small-scale exploratory study, demonstrate ChatGPT's potential in language education especially its capacity to deliver immediate feedback, promote personalization, foster critical thinking, and enhance autonomy. ChatGPT can provide personalized guidance, adapt to learners' language levels, engage them in interactive dialogue, and generate customized content tailored to students' needs. Its outputs often exhibit reasoned judgments, which can help mitigate potential threats to academic integrity; through immediate feedback, it promotes the development of critical thinking and autonomy.

Iqbal et al. (2025), through an exploratory sequential mixed-methods design, identify multiple advantages of GenAI in second language acquisition. It not only delivers highly personalized learning materials but also provides immediate feedback via interactive exercises and adaptive environments, significantly boosting learners'

motivation and engagement. The study highlights that integrating GenAI technologies can optimize teaching efficiency and support the creation of personalized learning pathways, thereby enabling more precise and effective second language instruction.

Bond et al. (2024) conduct a meta-review of reviews and systematically synthesize prior assessments of GenAI's application in education. Their findings show that adaptive learning systems exemplified by chatbots significantly enhance autonomy and critical thinking by personalizing learning content to meet individual needs. Furthermore, detailed immediate feedback mechanisms effectively improve writing quality and overall academic performance.

Rugaiyah (2023), in a systematic review, demonstrates that GenAI-driven language learning tools can substantially expand learners' vocabulary and grammatical knowledge, improve comprehension, and strengthen speaking skills through personalization, adaptive pathways, and immediate feedback. Whether by flagging grammatical errors, using conversational chatbots, or employing speech-recognition-based pronunciation correction, these tools help learners quickly detect and correct errors, thereby improving linguistic accuracy and fluency. Additionally, they track learning progress in real time and dynamically adjust difficulty and content, allowing learners to advance at their own pace and maximize learning effectiveness.

Hong (2023), in a literature review, reports that students can receive highly personalized tutoring via ChatGPT. This system generates discussion topics and creative writing prompts and effectively fosters development across the four language skills listening, speaking, reading, and writing while improving grammatical accuracy, expanding vocabulary, and enhancing learner motivation and confidence. ChatGPT can automatically correct essays, provide improvement suggestions, design lesson plans, and generate exercises and situational dialogues tailored to individual needs; immediate feedback plays a crucial role in this personalized learning experience.

Mendoza (2024), in a qualitative study, argues that GenAI texts can provide language learners with authentic, varied, and highly personalized input. Integrating

approaches such as critical digital pedagogy into language instruction is fundamental for constructing a pedagogical paradigm centered on critical thinking.

Al-Melhes (2023), in a practical SLR, explores AI's immediate feedback mechanisms and offers guidance on designing diverse feedback modalities to enhance autonomous learning on both mobile devices and computers.

Chicaíza et al. (2023), using a quantitative, non-experimental, cross-sectional design, show that AI can accurately identify learners' areas for improvement and provide precise personalized feedback, dynamically adjusting content. GenAI not only simulates native-speaker conversations and offers on-demand practice but also corrects grammatical errors, optimizes vocabulary use, and improves pronunciation through an interactive adaptive model that allows students to progress at their own pace.

Cisneros Vásquez et al. (2024), in a bibliographic review, conclude that GenAI's adaptive personalization and immediate feedback capabilities hold great potential to enhance educational outcomes; however, effectiveness depends greatly on system design, implementation, and management within educational contexts.

Ghafar et al. (2023) describe methods of data collection, focused scientific writing, literature review, and problem analysis, and explain how GenAI can support learner autonomy by continuously providing personalized instruction and immediate, scaffolded feedback.

Liu (2023), in a literature review, predicts that GenAI's future development in foreign language teaching will be broad and innovative. By leveraging speech recognition and adaptive technologies, GenAI systems will rapidly and accurately adjust instructional content to individual learner needs, enabling true personalization. Error correction and immediate feedback will remain essential for promoting learners' self-revision and autonomy.

Menacho et al. (2024), employing a quantitative design, empirically verify that systematically introducing GenAI technologies into classroom environments can guide

students toward autonomy and foster collaboration. Through intelligent platforms, this model dynamically tailors content to individual needs; automated assessment and immediate feedback support collaborative discussion and promote autonomy in higher education contexts.

Yunina (2023), using a literature-based approach, argues that GenAI provides personalized learning methods emphasizing autonomy and better outcomes in foreign language acquisition. AI-driven automated scoring applications, leveraging NLP and deep learning, deliver immediate feedback that supports individualized learning paths.

Zou et al. (2023), combining questionnaires, semi-structured interviews, and pre-/post-test data, demonstrate the motivational effect of social network-driven interaction on GenAI-assisted speaking practice. Learners maintain high engagement and receive personalized corrections in pronunciation, fluency, and lexical accuracy, resulting in significant speaking improvements.

Wexell Machado & Canese (2024), in a survey-based study, conclude that GenAI can personalize content and adjust pedagogical strategies to meet student needs. The immediate feedback and personalization provided by GenAI allow teachers to devote more time to complex, creative tasks while enhancing learner efficiency and motivation.

Gibert et al. (2024), using an exploratory design, demonstrate how GenAI tools can enhance language competencies by offering adaptive, personalized learning experiences. Strategically integrating technology into instructional design enables more accessible, personalized, and effective language teaching. Within personalized instruction frameworks, ChatGPT can stimulate thinking, refine ideas, provide immediate feedback, and promote self-assessment, thereby fostering deeper learning and consolidating knowledge.

Karataş et al. (2024), in a qualitative case study, demonstrate that ChatGPT plays a significant role in developing learners' critical thinking, providing immediate feedback, and fostering language skill development. It is particularly effective in

enhancing critical thinking, communication skills, and intercultural competence, and it shows substantial potential for offering personalized learning experiences and promoting collaboration.

Chan & Hu (2023), via a questionnaire, find that learners consider GenAI to have great potential for providing immediate feedback and tailored learning support, which can revolutionize traditional teaching methods and improve the efficiency of personalized learning.

Sasikumar & Sunil (2023), using AHP-TOPSIS multi-criteria decision analysis, reveal how GenAI tools through personalized interactions and immediate feedback help students generate and revise writing drafts, driving a shift toward dialogic and autonomous learning.

Hasyim et al. (2025), with a mixed-methods design investigating GenAI's role in fostering learner autonomy, find that GenAI holds considerable promise for personalizing instruction and promoting independence. Personalized, scaffolded feedback yields greater gains in learner motivation and facilitates faster progress in language skills than generic feedback.

Wei (2023), using a convergent mixed-methods approach, similarly reports transformative effects of AI platforms on engagement and personalized learning, showing that tailored practice and real-time corrective feedback enable learners to set individualized goals and monitor progress, thereby strengthening autonomy and confidence.

Qiao & Zhao (2023), using a pre-test/post-test design, provide empirical evidence that GenAI-mediated instruction benefits oral proficiency and self-regulation. Conversational agents can generate customized feedback and practice materials, enabling adaptive learning trajectories and effective scaffolding.

Suh et al. (2025), employing quantitative experimental methods, demonstrate that L2 learners can enhance critical thinking through interaction with ChatGPT. As a multifunctional conversational partner, ChatGPT supplies examples and writing

suggestions while prompting learners to question assumptions, challenge arguments, and evaluate evidence actions that foster deeper reflective reasoning.

Yatani et al. (2024), adopting a literature-based conceptual framework, argue that “externalizing-cognition” GenAI designed to provoke user cognition rather than simply supplying answers can promote critical thinking by posing reflective questions. Simulation-based designs should present diverse perspectives and include targeted scaffolding with planned fading to support internalization.

Law (2024), in a literature review, synthesizes evidence on the pedagogical value of GenAI in language classrooms, concluding that such tools can support acquisition, enhance skills, and deliver highly personalized learning experiences. GenAI’s roles as virtual tutor, expert advisor, learning partner, and reflective prompt can stimulate critical thinking and metacognitive practice.

Ogunleye et al. (2024), in a literature review, argue that GenAI offers scalable and affordable pathways to personalized learning. Their synthesis indicates that GenAI tools can generate tailored trajectories, elevate engagement, and support real-time interaction and feedback particularly through chatbots and adaptive response systems.

Roe & Perkins (2024), in a comprehensive review, find that diversified instructional materials combined with scalable personalized supports enhance the learner experience. They emphasize that conversational agents like ChatGPT can provide timely, sustained feedback in autonomous learning contexts, broadening access to instructional resources and supporting individualized pathways.

5. DISCUSSION

GenAI’s contribution to personalization is strongly supported by the reviewed literature. The analysis confirms that by producing graded, comprehensible, and interest-aligned input, it significantly expands opportunities for intake and incidental acquisition (Mouta et al., 2025; Son et al., 2025; Fernández, 2024). This aligns directly

with Krashen’s Input Hypothesis, as GenAI can generate abundant level-appropriate materials, and Tomlinson’s differentiated-instruction framework, since varying presentation, task complexity, and affective hooks increases engagement (Gutiérrez et al., 2025; Iqbal et al., 2025). Crucially, GenAI lowers the transaction costs of producing multiple leveled exemplars, enabling teachers to operationalize formative-assessment cycles that diagnose misunderstandings early and deliver diagnostic, actionable feedback tailored to each learner’s trajectory (Wexell Machado & Canese, 2024; Gibert et al., 2024). As a transition, these robust personalization affordances, widely documented across study methodologies, make downstream processes such as learner autonomy and scaffolded practice both more feasible and more effective when embedded in intentional instructional designs.

The claim regarding enhanced autonomy is substantiated by the evidence: GenAI effectively scaffolds self-regulated learning processes while also serving as a proximal mediator in sociocultural terms. From an SRL perspective, GenAI tools scaffold goal setting, strategy selection, metacognitive monitoring, and progress tracking by providing just-in-time prompts, exemplars, and reflective questions (Wei, 2023; Qiao & Zhao, 2023). From a Vygotskian standpoint, they act as a proximal scaffold within the ZPD, delivering contingent hints and modeled reformulations that are gradually faded as strategies are internalized (Mendoza, 2023; Menacho et al., 2024). This dual account, observed in both qualitative and quantitative studies, clarifies why autonomy increases when GenAI interactions are deliberately structured to encourage reflection and strategy transfer rather than unconditional answer-giving (Hasyim et al., 2025). It also underscores that genuine autonomy requires algorithmic literacy and teacher-facilitated micro-decisions (Al-Melhes, 2023; Liu, 2023). Transitioning from autonomy to feedback, these scaffolding functions set the stage for productive corrective cycles because learners who can monitor and enact strategies benefit more from immediate, targeted feedback.

The immediacy-of-feedback dimension emerges as the most prominently documented and impactful affordance in the reviewed literature. The evidence strongly

supports the proposition that instant, targeted corrective feedback promotes noticing and pushed output, thereby accelerating interlanguage development (Manzanedo, 2025; Rugaiyah, 2023; Hong, 2023). This is consistent with the Noticing Hypothesis (Schmidt) and the Output Hypothesis (Swain): instant corrective feedback and targeted reformulations make learner errors salient, prompting noticing, hypothesis testing, and interlanguage restructuring (Chicaíza et al., 2023; Zou et al., 2023). Furthermore, theories of skill acquisition suggest that frequent, low-stakes corrective cycles can accelerate the transfer from declarative knowledge to fluent procedural use, provided feedback is timely, specific, and paired with immediate reapplication (Chan & Hu, 2023; Sasikumar & Sunil, 2023). The reviewed studies, particularly the quantitative and mixed-methods research, echo that feedback's pedagogical value depends on its integration within the learning cycle rather than on speed alone (Bond et al., 2024; Ghafar et al., 2023). Moving toward higher-order outcomes, the combination of timely feedback and scaffolded reapplication also creates conditions in which GenAI can help cultivate metalinguistic awareness and critical thinking.

The development of critical-thinking skills through GenAI is confirmed, though the evidence base is more emergent. GenAI can function as a dialogic partner that elicits argumentative moves, alternative perspectives, and evidence evaluation, thereby supporting higher-order cognition (Karataş et al., 2024; Suh et al., 2025). When tasks require generation of counterclaims, critique, or synthesis, GenAI prompts argumentative strategies and evidence appraisal central to academic literacy (Law, 2024; Yatani et al., 2024; Mendoza, 2024). However, the cultivation of genuine critical thinking depends on attitudinal and metacognitive dispositions (e.g., cognitive flexibility, willingness to self-correct). Therefore, as highlighted across reviews and qualitative studies, AI-generated prompts must be embedded within pedagogies that explicitly train and assess these higher-order skills (Gutiérrez et al., 2025; Xiao & Zhi, 2023). This underscores that GenAI's dialogic affordances are necessary but not sufficient for durable gains in critical thinking unless combined with explicit instruction and assessment.

Taken together, the scoping review provides strong, multi-dimensional evidence supporting a coherent account: GenAI supplies comprehensible, scaffolded input (personalization), operates as a contingent mediator within the ZPD (autonomy), drives noticing and pushed output via immediacy-of-feedback, and functions as a dialogic tool that stimulates metacognition and critical engagement. The explanation refined by the evidence is that these mechanisms are deeply interlinked. For instance, personalization increases the relevance of input, which sustains autonomy, while immediate feedback relies on and reinforces the self-regulatory behaviors developed through autonomous practice. The combined effect of these dimensions depends on critical boundary conditions and generated materials must be contextually authentic; teacher mediation is essential to convert external scaffolds into durable internal strategies (a finding stressed across multiple reviews); and feedback alone is insufficient without learner engagement in deliberate reformulation. The transition to practice, therefore, calls for instructional designs that embed these four GenAI affordances within teacher-guided cycles of enactment, reflection, and planned fading of support, while concurrently cultivating learners' algorithmic literacy and strategic self-regulation.

6. CONCLUSION

The findings should be interpreted with caution because this review lacks primary classroom-based data, focuses mainly on ChatGPT, involves subjective literature selection, and only searched English and Spanish sources. Future studies should include multilingual searches, a broader range of GenAI tools, and empirical classroom data. This study conducted a scoping literature review to systematically examine the pedagogical potential of GenAI in second language teaching. The scoping literature review adhered to PRISMA 2020 reporting standards and was organized into three sequential phases: planning, execution, and reporting (Arksey & O'Malley, 2005; Peters et al., 2015). In the planning phase, I formulated the central research question: In which specific dimensions can GenAI most significantly support

the teaching and learning of a second language? I developed multiple search strings, which were executed across three databases: Google Scholar, Scopus, and Dialnet, yielding 1123 references. After two rounds of title/abstract and full-text screening, 74 articles satisfied the predefined inclusion criteria. To ensure source relevance and reliability, I applied a Likert-scale relevance rating and a double-blind review process. Ultimately, 35 studies were selected for in-depth analysis.

The findings reveal that ChatGPT offers substantial value in second language instruction across four interrelated dimensions: personalization, autonomy, immediate feedback, and critical thinking (Mouta et al., 2025; Son et al., 2025; Fernández, 2024). By continuously assessing learners' proficiency, interests, and goals, ChatGPT dynamically adapts task difficulty and exercise types, identifies recurrent errors, and provides targeted review questions or supplementary resources (Gutiérrez et al., 2025; Iqbal et al., 2025). These capabilities enhance both instructional efficiency and learner motivation (Wexell Machado & Canese, 2024; Gibert et al., 2024). The platform enables learners to engage actively at any time, request explanations or demonstrations, and exert fine-grained control over their learning pace and trajectory (Wei, 2023; Mendoza, 2023). Beyond delivering declarative knowledge, ChatGPT scaffolds self-assessment and reflective practice (Hasyim et al., 2025). Unlike traditional classroom feedback, which is often delayed, ChatGPT can detect grammatical, lexical, pragmatic, or logical errors in learners' written or spoken output and provide precise, actionable corrections immediately (Manzanedo, 2025; Rugaiyah, 2023). Furthermore, through open-ended prompts, role-play scenarios, and culturally diverse interlocutor simulations, ChatGPT promotes deeper analytical thinking, evaluative reasoning, and intercultural communicative competence (Karataş et al., 2024; Suh et al., 2025).

The study offers several actionable insights for educators and policymakers. First, educators can leverage ChatGPT to design personalized learning activities, foster learner autonomy, and integrate immediate feedback into classroom practice, thereby enhancing learning outcomes (Law, 2024; Yatani et al., 2024). Second,

instructors should cultivate strong critical evaluation skills and digital competencies to ensure the safe and effective integration of GenAI tools into their teaching, while simultaneously strengthening students' capacity to critically engage with GenAI during the learning process (Al-Melhes, 2023; Liu, 2023). Third, education administrators should establish clear guidelines addressing ethical AI use, data privacy, and academic integrity to guarantee the equitable and responsible deployment of GenAI in educational contexts (Gibert et al., 2024). Finally, by systematically identifying these four core dimensions, this study provides a conceptual framework for future empirical investigations into the application of diverse GenAI tools across different linguistic and cultural settings.

6.1. Gaps and contradictions in current research

Despite the burgeoning interest in integrating GenAI into language pedagogy, the body of literature addressing its specific applications in second language teaching and learning remains limited (Manzanedo, 2025; Al-Mughairi & Bhaskar, 2024; Mah & Groß, 2024). Existing studies reveal several notable gaps: the inherent challenges of incorporating body language, gestures, facial expressions, and affect have received insufficient attention; the processes by which AI-driven instructional materials are generated and designed are rarely described in detail; and the boundaries of what should be considered GenAI content remain ambiguous. Consequently, rigorous empirical research across diverse instructional contexts is needed to address these lacunae (Law, 2024; Son et al., 2025).

Gutiérrez et al. (2025) observe that ChatGPT's programming to avoid offending users leads it to adopt an overly positive or “benevolent” stance in nearly every interaction. This tendency may undermine the evaluation of learner output, particularly in written assignments or report writing, by focusing only on the most obvious errors while overlooking subtler issues such as paragraph length or lexical precision. Moreover, when ChatGPT generates illustrative graphics for exercises, these images

often lack clarity, impairing learners' comprehension or even rendering the tasks meaningless (Chan & Hu, 2023).

Although GPT models are trained on multilingual corpora, English-language texts and corresponding human supervision dominate the training data. In resource-scarce languages, the system frequently relies on machine translation, resulting in output that, while grammatically correct, remains structurally and stylistically aligned with English norms, thereby yielding uneven quality across languages (Mendoza, 2023).

In this study, ChatGPT, functioning as a GenAI tool, was found to significantly enhance four pivotal dimensions of second language instruction: personalization, autonomy, immediate feedback, and critical thinking. Nevertheless, dissenting perspectives caution against unreserved endorsement. Karataş et al. (2024) contend that although ChatGPT can foster reflective and critical thinking, it may concurrently impede learners' critical faculties and autonomy. The risk of generating misinformation and bias, along with ethical and privacy concerns, can foster academic dishonesty. Overreliance on such tools may erode fundamental learning skills and hinder the development of critical thinking and self-regulated learning. Learner apprehensions about these risks underscore the necessity of adopting a balanced integration strategy when incorporating ChatGPT and similar GenAI technologies into language curricula (Iqbal et al., 2025).

Mendoza (2024) further emphasizes that current GenAI conversational agents intended for blended foreign language classrooms cannot yet cater effectively to the specific needs of diverse learner populations or deliver genuinely personalized instruction. In many instances, these tools produce errors that neither second language learners nor heritage language learners can easily detect, thus contravening the core objectives of critical digital pedagogy.

Wexell Machado & Canese (2024) critique AI's potential to undermine the authenticity of instructional contexts and constrain creativity and critical inquiry. Similarly, Manzanedo (2025) finds that, despite ChatGPT's interactive learning

affordances, it still falls short in providing culturally situated oral communication and affective support.

Additional challenges include inadequate teacher training in GenAI applications, inequitable access to requisite technologies, and the limited adaptability of GenAI tools to varied cultural and linguistic groups (Changoluisa Santacruz et al., 2024). Learner concerns regarding ethics and privacy further highlight the exigency of robust regulatory and pedagogical frameworks to prevent plagiarism, ensure the integrity of student work, and develop effective GenAI integration strategies (Gibert et al., 2024).

Data security and system resilience constitute another critical risk area, given that GenAI platforms often store extensive learner data, which, if insufficiently protected, can lead to serious breaches (Yunina, 2023). Educators who integrate GenAI tools into their teaching must also navigate ethical challenges such as tool misuse and the facilitation of academic dishonesty, necessitating ongoing vigilance and regulation in practice (Mouta et al., 2025).

6.2. Future directions

To address the gaps and tensions, future research and practice must innovate pedagogically while steadfastly upholding principles of equity, responsibility, and transparency when deploying AI and virtualization tools in second language classrooms. First, learners' data privacy and security must be rigorously protected through robust protocols that prevent discriminatory or exclusionary access, ensuring that all students, regardless of socioeconomic status, cultural background, or technical proficiency, derive equitable benefits from these resources (Fernández, 2024).

Second, comprehensive ethical and social-impact frameworks should be established to help educators and institutions identify, mitigate, and monitor potential biases or exclusionary effects inherent in AI systems. Such frameworks ought to mandate clear attribution practices, requiring explicit labeling of which instructional or

assessment elements are GenAI, so as to distinguish human authorship from algorithmic output and preserve the distinct value of each (Mendoza, 2023).

The supervisory and corrective role of the teacher remains indispensable. Instructors should receive specialized training to critically evaluate, contextualize, and supplement GenAI feedback, thereby ensuring that learners receive accurate, constructive guidance that supports comprehensive language development, rather than uncritical “benevolent” praise. Professional development must also bolster educators’ digital literacy and ethical reasoning, empowering them to recognize AI “hallucinations”, address privacy concerns, and scaffold learners’ critical thinking skills. Collaborative co-design among technology developers, educational researchers, and classroom practitioners is vital. Pedagogically sound tools can be co-created to deliver intuitive, accessible solutions that integrate seamlessly into everyday instructional practice (Manzanedo, 2025).

There is an urgent need for large-scale, longitudinal, mixed-methods research to trace the evolution of AI integration across disciplines, age groups, and cultural environments. By combining quantitative surveys with in-depth interviews of teachers and stakeholders, and by developing refined assessment instruments that correlate specific modes of AI engagement with learning outcomes, researchers can identify the key factors influencing educators’ attitudes and practices. Such evidence will inform policy development, professional training design, and resource allocation, ensuring that AI-enhanced language learning remains inclusive, effective, and aligned with the humanistic core values of education (Iqbal et al., 2025; Chan & Hu, 2023; Ghimire et al., 2024).

Finally, AI is anticipated to deepen educational personalization by continuously analyzing individual learners’ trajectories and dynamically adapting both content and pedagogical strategies. Contextualized instruction will be enriched through authentic, game-based simulations, while the automation of routine administrative tasks, such as error logging and progress reporting, will liberate educators to concentrate on cultivating learners’ critical thinking and creative problem-solving abilities. An

integrated educational ecosystem, one that synergizes GenAI, project-based pedagogies, and collaborative learning platforms promises to render language acquisition ever more efficient, inclusive, and intrinsically motivating for all students learning a foreign language (Liu, 2023).

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